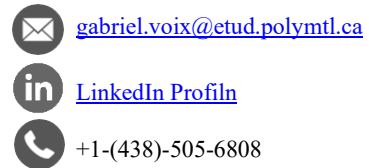


Gabriel Voix

Aerospace Engineering graduate

[Portfolio](#)



SUMMARY

Aerospace engineer graduated from Polytechnique Montreal, focused on composite structures, numerical simulation and manufacturing process, with industrial experiences at Airbus Toulouse and Bell Textron. Canadian and French citizen, eligible to work in Europe. Profile centered on the design - validation - manufacturing cycle, combining CAD, FEA, and advanced prototyping. Recognized for a strong ability to propose and implement technical solutions, rapidly assimilate new concepts, and drive projects end-to-end in an R&D environment.

EDUCATION

Bachelor's degree in Aerospace Engineering <i>École de Polytechnique Montréal, QC, Canada / With honors</i>	2022-2026 (grad. H2026)
Diploma of Collegial Studies in Natural Sciences <i>CÉGEP of St-Laurent, Montréal, QC, Canada</i>	2020-2022

EXPERIENCE

Filament winding lead for ORONOS	2023-Present
- Led the filament winding project from its inception in 2023 through the design and fabrication of the nose cone and inter-stage structures in 2025, culminating in breaking the Canadian student altitude record in 2025, with the next launch scheduled of a full filament wound rocket in 2026. - Validated the structural integrity of fuselages through destructive compression testing using Simcenter 3D simulations, laminate and failure criteria to analyze non-linear buckling behavior. - Develop a Python/PyQt software that generates winding path G-code to maximize the anisotropic properties of variable cross-section parts such as the nose cone, transition and COPV. - Write technical report sections on manufactured parts and train incoming members.	
Rapid Prototyping and Innovation Internship at Bell Textron	Apr-Sep 2025
- Simulate ply deformation under thermocompression using PAM-Composite for four Bell 505/407 panels, including one essential structural panel to identify critical shear and stress zones. - Performed ply layup and adjusted ply positioning in CATIA CPD based on simulation results, weekly presenting progress to the team in an iterative process. - Assemble and validate mold and ply geometry in CATIA to provide their exact 2D position to the laser projector on the assembly line and validate - Design, purchase, and assemble a robotic arm to scan and certify an electrical circuit sealing device on the Bell 429.	
International Internship at Airbus Operations SAS Laboratory	Apr-Sep 2024
- Design mechanical parts in CATIA for the cryogenic enclosure of a fatigue test bench used for composite testing of hydrogen COPV as part of the ZEROe project. - Develop a web application using the Python Django framework to build a database of 3,000 test laboratory tools, centralize equipment information, and provide a booking calendar. - Created technical drawings with GD&T specifications.	
Acoustic Research Assistant CRITIAS, École de Technologie Supérieure	Apr-Sep 2022
- Develop a Python program interfacing with the Zotero API to generate weekly reports on new scientific publications in acoustics. - Collaborate with engineers through continuous feedback to improve report quality.	

PROJECTS

COPV Design for Oronos

2026-Present

- Design a COPV with a 6061 aluminum liner overwrapped with carbon fiber by filament winding and tested the layup to reproduce the theoretical drape.
- Perform FEA on both the liner and composite to distribute stresses, minimize delamination zones, evaluate the autofrettage cycle, and optimize weight/pressure ratio in Simcenter 3D.
- Prepare machining paths and test procedures for weld and tubing leak tightness.

Design & Fabrication of a CNC Lathe, Senior Capstone Project

2025-Present

- Model a CNC lathe in SolidWorks meeting the design requirements and Canadian CSA standards.
- Design the electrical panel schematic, procured components, and assembled the circuit.
- Install LinuxCNC on the lathe and configure communication between all components.

Control System Design of a Thrust Vector Control Test Bench

Jan-May 2025

- Design the test bench to replicate the inertia and degrees of freedom of a rocket using a drone motor as the thrust source.
- Implement digital closed-loop control through iterative local optimization of PID gains until stabilization using an Arduino.

SKILLS

Software: NX, Simcenter 3D, PAM-Composite, CATIA V5, SolidWorks, Fusion360, ANSYS

Programing: Python (Django framework, PyQt), G-code, Matlab, GitHub, HTML, CSS, JavaScript

Complementary skills: CNC machining, laser cutting, 3D printing FDM/SLS/SLA

LANGUAGES

French – native language

English – bilingual